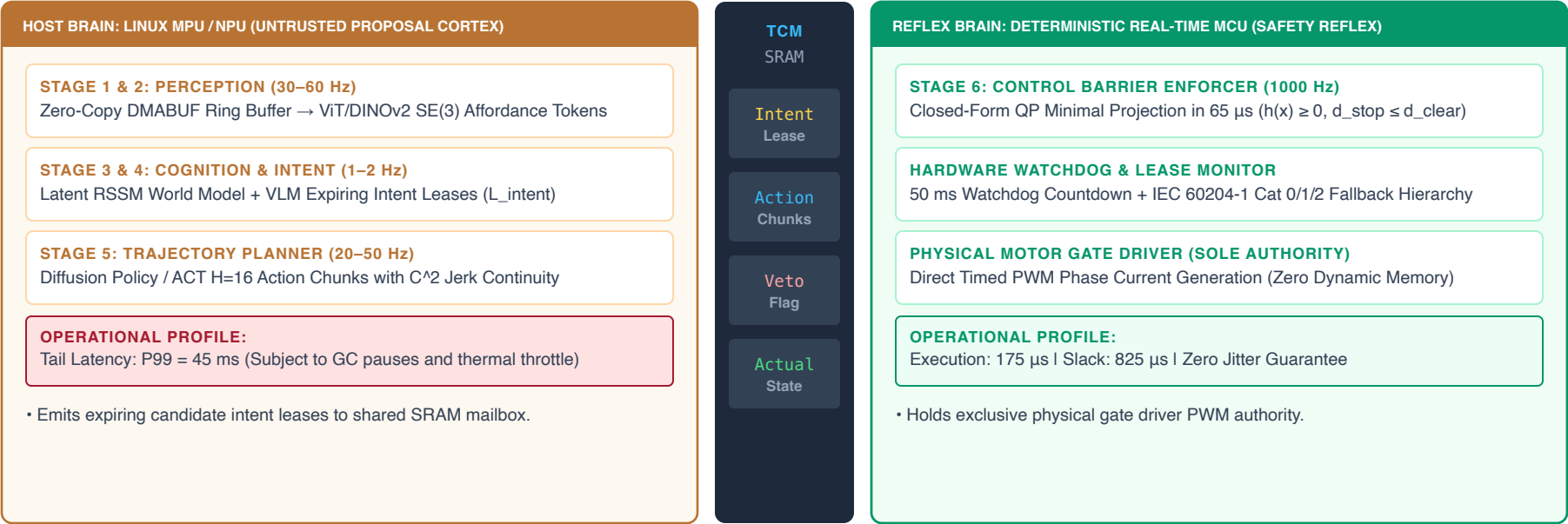


END-TO-END DUAL-BRAIN AGENT ARCHITECTURE ON ARDUINO UNO Q /PHYSICAL AI KIT

Hardware co-design: Linux MPU (Host Brain) proposal cortex and bare-metal MCU (Reflex Brain) safety reflex over shared SRAM



THE ARDUINO UNO Q DUAL-BRAIN ARCHITECTURAL CONTRACT

- Proposal Privilege Separation:** The Linux MPU runs high-capacity AI models as untrusted proposal generators.
- Deterministic Safety Invariants:** The Cortex-M4 MCU evaluates mathematical safety barriers at 1000 Hz, with zero OS or heap dependencies.
- Panic Survivability:** Total Linux crash or NPU bus hang triggers unmaskable hardware fallback in < 35 ms without physical injury.